

AFTER MIDS

Topic:

Total Derivative in Chain Rule

Exercise: NO: 14.4 (Q1, to Q6)

Chain Rule: One Independent Variable

Question: No: 01

$w = x^2 + y^2$; $x = \cos t$, $y = \sin t$,
 $t = \pi$. Find $\frac{dw}{dt}$.

Solution:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

$$w = x^2 + y^2$$

$$\frac{\partial w}{\partial x} = 2x$$

$$\frac{\partial w}{\partial y} = 2y$$

$$x = \cos t, \text{ so}$$

$$\frac{dx}{dt} = -\sin t$$

$$y = \sin t, \text{ so}$$

$$\frac{dy}{dt} = \cos t$$

Now,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}$$

Putting the values, so get

$$\frac{dw}{dt} = (2x)(-\sin t) + (2y)(\cos t)$$

$$\frac{dw}{dt} = -2x \sin t + 2y \cos t \quad \text{at } (t = \pi)$$

$$\frac{dw}{dt} = -2x \sin \pi + 2y \cos \pi$$

$$\frac{dw}{dt} = 0 - 2y$$

$$\frac{dw}{dt} = -2y \quad \text{Ans.} \quad \frac{dw}{dt} = -2(\sin t) = -2 \sin \pi$$

$$\frac{dw}{dt} = -2(0) \Rightarrow \frac{dw}{dt} = 0 \quad \text{Ans.}$$

Question: No: 02

$$w = x^2 + y^2; \quad x = \cos t + \sin t, \quad y = \cos t - \sin t$$

$$t = 0. \quad \text{Find } \frac{dw}{dt}$$

Solution:

$$w = x^2 + y^2$$

$$\frac{\partial w}{\partial x} = \cancel{\sin t} = \cos t \quad 2x$$

$$\frac{\partial w}{\partial y} = 2y$$

$$x = \cos t + \sin t$$

$$\frac{dx}{dt} = -\sin t + \cos t$$

$$y = \cos t - \sin t$$

$$\frac{dy}{dt} = -\sin t - \cos t$$

Now,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

Putting the values so get

$$\frac{dw}{dt} = (2x)(-\sin t + \cos t) + 2y(-\sin t - \cos t)$$

$$\frac{dw}{dt} = 2(\cos t + \sin t)(-\sin t + \cos t) +$$

$$2(\cos t - \sin t)[-(\cos t + \sin t)]$$

$$\frac{dw}{dt} = 2(\cos^2 t - \sin^2 t) + [-2(\cos^2 t - \sin^2 t)]$$

$$\frac{dw}{dt} = 2(\cos^2 t - \sin^2 t) - 2(\cos^2 t - \sin^2 t)$$

$$\frac{dw}{dt} = 0$$

$$\frac{dw}{dt}(0) = 0 \quad \text{Ans}$$

Question: Nos 03

$$w = \frac{x}{2} + \frac{y}{2}, \quad x = \cos^2 t, \quad y = \sin^2 t$$

$$z = \frac{1}{t}; \quad t = 3. \quad \text{Find } \frac{dw}{dt}$$

Solution:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Putting the values, so get

$$\frac{dw}{dt} = \frac{\partial}{\partial x} \left(\frac{x}{2} + \frac{y}{2} \right) \cdot \frac{d}{dt} (\cos^2 t) + \frac{\partial}{\partial y} \left(\frac{x}{2} + \frac{y}{2} \right) \cdot \frac{d}{dt} (\sin^2 t) + \frac{\partial}{\partial z} \left(\frac{x}{2} + \frac{y}{2} \right) \cdot \frac{d}{dt} \left(\frac{1}{t} \right)$$

$$\frac{dw}{dt} = \left(\frac{1}{2}\right)(-2 \cos t \sin t) + \left(\frac{1}{2}\right)(2 \sin t \cos t) + \left(\frac{-x}{z^2} - \frac{y}{z^2}\right)\left(-\frac{1}{t^2}\right)$$

$$\frac{dw}{dt} = -\frac{2}{2}(\cos t \sin t) + \frac{2}{2} \sin t \cos t - \left(\frac{x+y}{z^2}\right)\left(-\frac{1}{t^2}\right)$$

$$\frac{dw}{dt} = \frac{x+y}{z^2 t^2}$$

Putting the values, so

$$\frac{dw}{dt} = \frac{\cos^2 t + \sin^2 t}{(1/4)^2 (t)^2} \quad \because (\sin^2 t + \cos^2 t = 1)$$

$$\frac{dw}{dt} = \frac{1}{\frac{1}{16} \cdot 4^2} = \frac{1}{1} = 1$$

$$\frac{dw}{dt}(3) = 1 \text{ Ans.}$$

Question: Nos 04

$$w = \ln(x^2 + y^2 + z^2), \quad x = \cos t, \quad y = \sin t$$

$$z = 4\sqrt{t} \quad ; \quad t = 3. \text{ Find } \frac{dw}{dt}.$$

Solution:

As we know that

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{\partial w}{\partial x} = \frac{2}{\partial x} [\ln(x^2 + y^2 + z^2)]$$

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}$$

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By chain rule

$$\ln(x^2 + y^2 + z^2) = \ln(u), \text{ so}$$

$$\frac{d}{du} [f(g(u))] = f'(g(u)) \cdot g'(u)$$

Now,

$$f(u) = \ln(u) \quad f'(u) = \frac{1}{u}$$

And

$$g(u) = (x^2 + y^2 + z^2) \quad \frac{d}{du} = 2u$$

Then:

$$\frac{d}{du} [f(g(u))] = f'(g(u)) \cdot g'(u)$$

$$= \frac{1}{u} \cdot 2u$$

$$\frac{\partial w}{\partial x} = \frac{2u}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial y} = \frac{\partial y}{\partial (x^2 + y^2 + z^2)} \quad (\text{By using upper method})$$

$$\frac{dx}{dt} = \frac{d}{dt} (\cos t) = -\sin t$$

$$\frac{dy}{dt} = \frac{d}{dt} (\sin t) = \cos t$$

$$\frac{\partial w}{\partial y} = \frac{\partial z}{\partial (x^2 + y^2 + z^2)} \quad (\text{By using } \frac{\partial w}{\partial u} \text{ method})$$

$$\frac{dz}{dt} = \frac{d}{dt} (4\sqrt{t})$$

$$= \frac{d}{dt} 4(t)^{1/2} = 4 \cdot \frac{1}{2} t^{-1/2} = \frac{4}{2} t^{-1/2}$$

By using power rule

$$\frac{dz}{dt} = \frac{4}{2\sqrt{t}}$$

Now,

$$\frac{dw}{dt} = \frac{\partial w}{\partial u} \cdot \frac{du}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Putting the values, so get

$$\frac{dw}{dt} = \frac{\partial w}{\partial u} (-\sin t) + \frac{\partial w}{\partial y} (\cos t) +$$

$$\frac{\partial w}{\partial z} \left(\frac{4}{2\sqrt{t}} \right)$$

Put the value of u, y and z , so

$$\frac{dw}{dt} = \frac{2}{(x^2 + y^2 + z^2)} (-\cos t \sin t) + \frac{2}{(x^2 + y^2 + z^2)} (\sin t \cos t)$$

$$+ \frac{2}{(x^2 + y^2 + z^2)} \left(\frac{4}{2\sqrt{t}} \right)$$

$$\frac{dw}{dt} = \frac{-2 \cos t \sin t}{x^2 + y^2 + z^2} + \frac{2 \sin t \cos t}{x^2 + y^2 + z^2}$$

$$+ \frac{2}{x^2 + y^2 + z^2} (8)$$

$$\frac{dw}{dt} = \frac{16}{x^2 + y^2 + z^2} \quad (\text{Putting the value of } x, y, z)$$

$$\frac{dw}{dt} = \frac{16}{\cos^2 t + \sin^2 t + (4\sqrt{t})^2} \quad \text{as } (\cos^2 t + \sin^2 t = 1)$$

$$\frac{dw}{dt} = \frac{16}{1 + 16t}$$

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$$\frac{dw}{dt}(3) = \frac{16}{1+16(3)} \quad \text{or } t=3$$

$$\frac{dw}{dt} = \frac{16}{49} \quad \text{Ans.}$$

Question: NO: 05

$$w = 2ye^x - \ln z, \quad x = \ln(t^2 + 1),$$

$$y = \tan^{-1} t, \quad z = e^t : t = 1. \text{ Find } \frac{dw}{dt}.$$

Solution:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{dw}{dt} = \frac{\partial}{\partial x} (2ye^x - \ln z) \left(\frac{d}{dt} (\ln(t^2 + 1)) \right) +$$

$$\frac{\partial}{\partial y} (2ye^x - \ln z) \left(\frac{d}{dt} (\tan^{-1} t) \right) + \left(\frac{\partial}{\partial z} (2ye^x - \ln z) \right) \cdot$$

$$\cdot \left(\frac{d}{dt} (e^t) \right)$$

$$\frac{dw}{dt} = 2ye^x \cdot \frac{2t}{t^2 + 1} + 2e^x \cdot \frac{1}{1+t^2} + \frac{-1}{z} \cdot e^t$$

$$\frac{dw}{dt} = \frac{4 \tan^{-1} t \cdot e^{\ln(t^2 + 1)}}{(t^2 + 1)} + \frac{2e^{\ln(t^2 + 1)}}{t^2 + 1} - \frac{e^t}{e^t}$$

$$\frac{dw}{dt} = \frac{4 (\tan^{-1} 1) (t^2 + 1)}{(t^2 + 1)} + \frac{2 (t^2 + 1)}{(t^2 + 1)} - 1$$

$$\frac{dw}{dt} = 4 (\tan^{-1} 1) 1 + 2 - 1$$

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$$\frac{dw}{dt} = \frac{4(\pi)}{4} + 1$$

$$\frac{dw}{dt} = (\pi + 1) \text{ OR } 180 + 1 = 181 \text{ Ans.}$$

Question: No. 06

$w = z - \sin xy$, $x = t$, $y = \ln t$,
 $z = e^{t-1}$; $t = 1$. Find $\frac{dw}{dt}$.

Solution:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

$$\frac{dw}{dt} = \frac{\partial}{\partial x} (z - \sin xy) \cdot \frac{d}{dt} (t) + \frac{\partial}{\partial y} (z - \sin xy) \cdot \frac{d}{dt} (\ln t) + \frac{\partial}{\partial z} (z - \sin xy) \cdot \frac{d}{dt} (e^{t-1})$$

$$\frac{dw}{dt} = \frac{d}{dt} (\ln t) + \frac{\partial}{\partial z} (z - \sin xy) \cdot \frac{d}{dt} (e^{t-1})$$

$$\frac{dw}{dt} = -y \cos xy (1) + (-x \cos xy) \left(\frac{1}{t}\right) +$$

$$1) (e^{t-1})$$

$$\frac{dw}{dt} = -\ln t [\cos t \cdot \ln t] - \frac{t \cos t \cdot \ln t}{t} - e^{t-1}$$

$$\frac{dw}{dt} = -\ln 1 [\cos 1 \cdot \ln 1] - (\cos 1 \cdot \ln 1) + e^{1-1}$$

$$\frac{dw}{dt} = 0 - 1 + e^0 \quad \because (e^0 = 1)$$

$$\frac{dw}{dt} = -1 + 1 = 0 \text{ Ans.}$$

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Thomas Calculus (E-11)

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Exercise NO: 15.2

Topic:

Multiply Integration

 $\Rightarrow$ Double Integrals $\Rightarrow$ Find the integration and double integrals.

Questions NO: 01

$$\int_0^3 \int_0^2 (4 - y^2) dy dx$$

Solution:

$$= \int_0^3 \int_0^2 (4 - y^2) dy dx$$

$$= \int_0^3 \left[4y - \frac{y^3}{3} \right]_0^2 dx$$

$$= \int_0^3 \left[4(2) - \frac{(2)^3}{3} \right] - \left[4(0) - \frac{(0)^3}{3} \right] dx$$

$$= \int_0^3 \left(8 - \frac{8}{3} \right) dx$$

$$= \int_0^3 \left(\frac{24-8}{3} \right) dx = \int_0^3 \left(\frac{16}{3} \right) dx$$

$$= \frac{16}{3} \int_0^3 1 dx$$

$$= \frac{16}{3} \left[x \right]_0^3 + C$$

$$= \frac{16}{3} [3 - 0] + C$$

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$$= \frac{16}{3}$$

Solution

$$= \int_0^3$$

$$= \int_0^3$$

$$= \int_0^3$$

$$= \int_0^3 [x^2$$

$$= \int_0^3 0 -$$

$$= - \int_0^3$$

$$= - \int_0^3 \frac{1}{x^2}$$

$$= - \left[\frac{2(3)}{3} \right]$$

$$= - [2(27)]$$

$$= - \frac{3}{18}$$

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$$= \frac{16}{3} (3) + C \Rightarrow = 16 + C \text{ Ans.}$$

Question: NO: 02

$$\int_0^3 \int_{-2}^0 (x^2 y - 2xy) dy dx$$

Solution:

$$= \int_0^3 \int_{-2}^0 (x^2 y - 2xy) dy dx$$

$$= \int_0^3 \left[\frac{x^2 y^2}{2} - \frac{2x y^2}{2} \right]_{-2}^0 dx$$

$$= \int_0^3 \left[\frac{x^2 y^2}{2} - x(y)^2 \right]_{-2}^0 dx$$

$$= \int_0^3 \left[\frac{x^2 (0)^2}{2} - x(0)^2 \right] - \left[\frac{x^2 (-2)^2}{2} - x(-2)^2 \right] dx$$

$$= \int_0^3 0 - \left[\frac{4x^2}{2} - 4x \right] dx$$

$$= \int_0^3 2x^2 - 4x dx$$

$$= \left[\frac{2x^3}{3} - \frac{4x^2}{2} \right]_0^3 + C$$

$$= \left[\frac{2(3)^3}{3} - \frac{4(3)^2}{2} \right] - \left[\frac{2(0)^3}{3} - \frac{4(0)^2}{2} \right] + C$$

$$= \left[\frac{2(27)}{3} - \frac{4(9)}{2} \right] - 0$$

$$= 18 - 18$$

$$= 0 \text{ Ans.}$$

Question: NO: 03

$$\int_{-1}^0 \int_{-1}^1 (x+y+1) dx dy$$

Solution:

$$= \int_{-1}^0 \left[\frac{x^2}{2} + xy + x \right]_{-1}^1 dy$$

$$= \int_{-1}^0 \left[\frac{(1)^2}{2} + (1)y + (1) \right] - \left[\frac{(-1)^2}{2} + (-1)y + (-1) \right] dy$$

$$= \int_{-1}^0 \left(\frac{1}{2} + y + 1 - \frac{1}{2} + y + 1 \right) dy$$

$$= \int_{-1}^0 (2y + 2) dy$$

$$= \left[\frac{2y^2}{2} + 2y \right]_{-1}^0 + C$$

$$= \left[\frac{2(0)^2}{2} + 2(0) \right] - \left[\frac{2(-1)^2}{2} + 2(-1) \right] + C$$

$$= 0 - (+1 - 2) + C$$

$$= -(-1) + C$$

$$= 1 + C \text{ Ans.}$$

Question: NO: 04

$$\int_{\pi}^{2\pi} \int_0^{\pi} (\sin x + \cos y) dx dy$$

Solution:

$$= \int_{\pi}^{2\pi} \left[\sin x + x \cos y \right]_0^{\pi} dy$$

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$$= \int_{\pi}^{2\pi} [-\cos \pi + \pi \cos y] - [-\cos(0) + 0 \cos y] dy$$

$$= \int_{\pi}^{2\pi} (-\cos \pi + \pi \cos y) dy$$

$$= \int_{\pi}^{2\pi} (-\cos \pi + \pi \cos y) dy$$

$$= \int_{\pi}^{2\pi} (-(-1) + 180 \cos y + 1) dy$$

$$= \int_{\pi}^{2\pi} (1 + \pi \cos y + 1) dy$$

$$= \int_{\pi}^{2\pi} (2 + \pi \cos y) dy$$

$$= \left[2y + \pi \sin y \right]_{\pi}^{2\pi} + C$$

$$= [2(2\pi) + \pi \sin(2\pi)] - [2(\pi) + \pi \sin(\pi)] + C$$

$$= [4\pi + \pi(0)] - [2\pi + \pi(0)] + C$$

$$= 4\pi - 2\pi$$

$$= 2\pi \text{ Ans.}$$

Question NO: 05

$$\int_0^{\pi} \int_0^x x \sin y \, dy \, dx$$

Solution:

$$= \int_0^{\pi} \int_0^x x \sin y \, dy \, dx$$

By using integration
By parts.

$$= \int_0^{\pi} \left[x(-\cos y) \right]_0^x dx$$

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$$= \int_0^\pi \left| -u \cos u \right|_0^u du$$

$$= \int_0^\pi (-u \cos u) - (-u \cos 0) du$$

$$= \int_0^\pi -u \cos u - (-u(1)) du$$

$$= \int_0^\pi u - u \cos u du \rightarrow \text{By using integration by part, so}$$

$$= \left[\frac{u^2}{2} - (u \sin u + \cos u) \right]_0^\pi$$

$$= \left[\frac{\pi^2}{2} - (\pi \sin \pi + \cos \pi) \right] - \left[\frac{0^2}{2} - (0 \sin 0 + \cos 0) \right]$$

$$= \frac{\pi^2}{2} - [\pi(0) + (-1)] - \{ 0 - [0(0) + 1] \}$$

$$= \frac{\pi^2}{2} + 1 + 1$$

$$= \frac{\pi^2}{2} + 2 + C \text{ Ans.}$$

Question: NO: 06

$$\int_0^\pi \int_0^{\sin u} y \, dy \, du$$

Solution:

$$= \int_0^\pi \left[\frac{y^2}{2} \right]_0^{\sin u} du$$

$$= \int_0^\pi \left[\frac{(\sin u)^2}{2} \right] - \left[\frac{(0)^2}{2} \right] du$$

$$= \int_0^\pi \frac{1}{2} \sin^2 u \, du$$

$$= \frac{1}{2} \int_0^\pi \sin^2 u \, du$$

$$\because \sin^2 u = \frac{1 - \cos 2u}{2}$$

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$$= \frac{1}{2} \int_0^\pi$$

$$= \frac{1}{4} \int_0^\pi$$

$$= \frac{1}{4} \left[\int_0^\pi$$

$$= \frac{1}{4} \left[\pi$$

$$= \frac{1}{4} \left[\pi$$

$$= \frac{1}{4} \left(\pi$$

$$= \frac{1}{4}$$

$$\int_1^{\ln 8}$$

Solution

$$= \int_1^{\ln 8}$$

$$= \int_2^{\ln 8} \left[e$$

$$= \int_1^{\ln 8}$$

$$= \int_1^{\ln 8} (e$$

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$$= \frac{1}{2} \int_0^{\pi} \frac{1 - \cos 2u}{2} du$$

$$= \frac{1}{4} \int_0^{\pi} 1 - \cos 2u du$$

$$= \frac{1}{4} \left[\int_0^{\pi} 1 du - \int_0^{\pi} \cos 2u du \right]$$

$$= \frac{1}{4} \left[u - \frac{\sin 2u}{2} \right]_0^{\pi}$$

$$= \frac{1}{4} \left[\left[\pi - \frac{\sin 2(\pi)}{2} \right] - \left[0 - \frac{\sin 2(0)}{2} \right] \right]$$

$$= \frac{1}{4} (\pi - 0) - 0$$

$$= \frac{1\pi}{4} + c \text{ Ans.}$$

Question: NO: 07

$$\int_1^{\ln 8} \int_0^{\ln y} e^{x+y} dx dy$$

Solution:

$$= \int_1^{\ln 8} \left[e^{x+y} \right]_0^{\ln y} dy$$

$$= \int_1^{\ln 8} [e^{\ln y + y} - e^{0+y}] dy$$

$$= \int_1^{\ln 8} (e^{\ln y} \cdot e^y - e^0 \cdot e^y) dy \quad \because e^0 = 1$$

$$= \int_1^{\ln 8} (e^{\ln y} \cdot e^y - e^y) dy$$

$$= \int_1^{\ln 8} (ye^y - e^y) dy$$

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$$= \int_1^{\ln 8} y e^y dy - \int_1^{\ln 8} e^y dy$$

Applying
integration
by parts

$$= \left[y e^y - \int_1^{\ln 8} \frac{d}{dy} (e^y \cdot y) dy \right] - \int_1^{\ln 8} e^y dy$$

$$= y e^y - \int_1^{\ln 8} e^y (1) dy - \int_1^{\ln 8} e^y dy$$

$$= \left[y e^y - e^y - e^y \right]_1^{\ln 8} = \left[y e^y - 2e^y \right]_1^{\ln 8}$$

$$= \left[\ln 8 e^{\ln 8} - 2e^{\ln 8} \right] - \left[1e^1 - 2e^1 \right]$$

$$= (8 \ln 8 - 16) - (-e^1)$$

$$= 8 \ln 8 - 16 + e + C \text{ Ans.}$$

Question: NO: 08

$$\int_1^2 \int_y^{y^2} dx dy$$

Solution:

$$= \int_1^2 \int_y^{y^2} dx dy$$

$$= \int_1^2 \left[x \right]_y^{y^2} dy$$

$$= \int_1^2 [y^2 - y] dy$$

$$= \int_1^2 y^2 - \int_1^2 y dy$$

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$$= \left[\frac{y^3}{3} - \frac{y^2}{2} \right]_1^2$$

$$= \left[\frac{(2)^3}{3} - \frac{(2)^2}{2} \right] - \left[\frac{(1)^3}{3} - \frac{(1)^2}{2} \right] + C$$

$$= \left[\frac{8}{3} - \frac{4}{2} \right] - \left[\frac{1}{3} - \frac{1}{2} \right] + C$$

$$= \left(\frac{16-12}{6} \right) - \left(\frac{2-3}{6} \right) + C$$

$$= \left(\frac{4}{6} \right) - \left(-\frac{1}{6} \right) + C$$

$$= \left(\frac{2}{3} + \frac{1}{6} \right) + C$$

$$= \frac{12+3}{18} + C \Rightarrow = \frac{15}{18} = \frac{5}{6} \text{ Ans.}$$

Question: NO: 09

$$\int_0^1 \int_0^{y^2} 3y^3 e^{xy} \, dx \, dy$$

Solution:

$$= \int_0^1 3 \int_0^{y^2} y^3 e^{xy} \, dx \, dy$$

$$= \int_0^1 3y^3 \int_0^{y^2} e^{xy} \, dx \, dy$$

$$= \int_0^1 \left[\frac{3y^3 \cdot e^{xy}}{y} \right]_0^{y^2} dy$$

$$= \int_0^1 \left[3y^2 e^{xy} \right]_0^{y^2} dy$$

$$= \int_0^1 \left[3y^2 e^{y^3 \cdot y} \right] - \left[3y^2 e^{0 \cdot y} \right] dy \quad \because e^0 = 1$$

$$= \int_0^1 (3y^2 e^{y^3} - 3y^2) dy$$

$$= \int_0^1 3y^2 e^{y^3} dy - \int_0^1 3y^2 dy$$

In this portion we use substitution

$$\text{Let } x = y^3$$

$$\frac{dx}{dy} = 3y^2$$

$$dx = 3y^2 dy, \text{ So}$$

$$= \int_0^1 e^x dx - \int_0^1 3y^2 dy$$

$$= \left[e^x - \frac{3y^3}{3} \right]_0^1 + C$$

$$= \left[e^x - y^3 \right]_0^1 \text{ again put the value of } x = y^3, \text{ So}$$

$$= \left[e^{y^3} - y^3 \right]_0^1 \Rightarrow \left[e^{(1)^3} - (1)^3 \right] - \left[e^{(0)^3} - (0)^3 \right]$$

$$= (e^1 - 1) - (e^0 - 0) + C$$

$$= (e^1 - 1) - (1 - 0) + C$$

$$= e - 1 - 1 + C$$

$$= e - 2 + C$$

Ans.

Solution

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Question: NO: 10

$$\int_1^4 \int_0^{\sqrt{u}} \frac{3}{2} e^{y/\sqrt{u}} dy du$$

Solution:

$$= \int_1^4 \int_0^{\sqrt{u}} \frac{3}{2} e^{y/\sqrt{u}} dy du$$

$$= \int_1^4 \frac{3}{2} \int_0^{\sqrt{u}} e^{y/\sqrt{u}} dy du$$

$$= \int_1^4 \left[\frac{3}{2} \cdot \frac{e^{y/\sqrt{u}}}{1/\sqrt{u}} \right]_0^{\sqrt{u}} du$$

$$= \int_1^4 \left[\frac{3}{2} e^{y/\sqrt{u}} (\sqrt{u}) \right]_0^{\sqrt{u}} du$$

$$= \frac{3}{2} \int_1^4 \left[e^{y/\sqrt{u}} (\sqrt{u}) \right]_0^{\sqrt{u}} du$$

$$= \frac{3}{2} \int_1^4 \sqrt{u} \left[e^{y/\sqrt{u}} \right]_0^{\sqrt{u}} du$$

$$= \frac{3}{2} \int_1^4 \sqrt{u} (e^{y/\sqrt{u}} - e^{0/\sqrt{u}}) du$$

$$\because e^0 = 1$$

$$= \frac{3}{2} \int_1^4 \sqrt{u} (e^1 - 1) du$$

$$= \frac{3}{2} (e^1 - 1) \int_1^4 \sqrt{u} du$$

$$= \frac{3}{2} (e - 1) \int_1^4 x^{1/2} du$$

$$= \frac{3}{2} (e - 1) \left[\frac{x^{3/2}}{3/2} \right]_1^4$$

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$$= \frac{7}{2} (e-1) \left[\frac{(4)^{3/2}}{3/2} - \frac{(1)^{3/2}}{3/2} \right]$$

$$= (e-1) ((2)^3 - 1)$$

$$= (e-1) (8 - 1)$$

$$= (e-1) (8 - 1)$$

$$= (e-1) 7$$

$$= 7 (e-1) \text{ Ans.}$$

Question: NO: 17

$$\int_{-2}^0 \int_v^{-v} 2 \, dp \, dv$$

Solution:

$$= \int_{-2}^0 2 \int_v^{-v} 1 \, dp \, dv$$

$$= \int_{-2}^0 2 \left[p \right]_v^{-v} dv$$

$$= \int_{-2}^0 2 [-v - v] dv$$

$$= \int_{-2}^0 -2v - 2v \, dv$$

$$= \int_{-2}^0 -4v \, dv$$

$$= -4 \int_{-2}^0 v \, dv$$

$$= -4 \left[\frac{v^2}{2} \right]_{-2}^0$$

$$= -4 \left[\frac{(0)^2}{2} - \frac{(-2)^2}{2} \right]$$

$$= -4 \left[0 - \frac{4}{2} \right]$$

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$$= -4(-2) \Rightarrow = 8 \text{ Ans.}$$

Question: NO: 18

$$\int_0^1 \int_0^{\sqrt{1-s^2}} 8t \, dt \, ds$$

Solution:

$$= \int_0^1 \int_0^{\sqrt{1-s^2}} 8t \, dt \, ds$$

$$= \int_0^1 8 \int_0^{\sqrt{1-s^2}} t \, dt \, ds$$

$$= \int_0^1 8 \left[\frac{t^2}{2} \right]_0^{\sqrt{1-s^2}} ds$$

$$= \int_0^1 8 \left[\frac{(\sqrt{1-s^2})^2}{2} - \frac{(0)^2}{2} \right] ds$$

$$= \int_0^1 8 \left(\frac{1-s^2}{2} \right) ds$$

$$= 4 \int_0^1 1-s^2 \, ds$$

$$= 4 \left[\int_0^1 1 \, ds - \int_0^1 s^2 \, ds \right]$$

$$= 4 \left[[s]_0^1 - \left[\frac{s^3}{3} \right]_0^1 \right]$$

$$= 4 \left[(1) - \frac{1}{3} \right]$$

$$= 4 \left[\frac{3-1}{3} \right] \Rightarrow = 4 \left(\frac{2}{3} \right)$$

$$= \frac{8}{3} \text{ Ans.}$$

Question: NO: 19

$$\int_{-\pi/3}^{\pi/3} \int_0^{\sec t} 3 \cos t \, du \, dt$$

Solution:

$$= \int_{-\pi/3}^{\pi/3} (3 \cos t) \int_0^{\sec t} 1 \, du \, dt$$

$$= \int_{-\pi/3}^{\pi/3} \left[(3 \cos t) u \right]_0^{\sec t} dt$$

$$= \int_{-\pi/3}^{\pi/3} (3 \cos t \sec t) dt$$

$$= \int_{-\pi/3}^{\pi/3} \left(3 \cos t \cdot \frac{1}{\cos t} \right) dt$$

$$= \int_{-\pi/3}^{\pi/3} 3 \, dt$$

$$= 3 \int_{-\pi/3}^{\pi/3} 1 \, dt \Rightarrow = \left[3t \right]_{-\pi/3}^{\pi/3}$$

$$= 3 \left[\frac{\pi}{3} - \left(-\frac{\pi}{3} \right) \right] \Rightarrow = 3 \left(\frac{\pi}{3} + \frac{\pi}{3} \right)$$

$$= 3 \left(\frac{\pi + \pi}{3} \right) \Rightarrow = 3 \left(\frac{2\pi}{3} \right)$$

$$= 2\pi \text{ Ans.}$$

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Question: No: 20

$$\int_0^3 \int_1^{4-2u} \frac{4-2u}{v^2} dv du$$

Solution:

$$= \int_0^3 \int_0^{4-2u} (4-2u) v^{-2} dv du$$

$$= \int_0^3 (4-2u) \int_0^{4-2u} v^{-2} dv du$$

$$= \int_0^3 \left[(4-2u) \cdot \frac{v^{-1}}{-1} \right]_0^{4-2u} du$$

$$= \int_0^3 \left[(4-2u) \cdot -v^{-1} \right]_0^{4-2u} du$$

$$= - \int_0^3 \left[\frac{(4-2u)}{v} \right]_0^{4-2u} du$$

$$= - \int_0^3 \left[\frac{-4+2u}{v} \right]_0^{4-2u} du$$

Putting the limits, so

$$= \int_0^3 \left(\frac{2u-4}{4-2u} \right) - \left(\frac{2u-4}{1} \right) du$$

$$= \int_0^3 \frac{2u-4}{-(2u-4)} - 2u+4 du$$

$$= \int_0^3 (-1 - 2u + 4) du$$

$$= \int_0^3 (3-2u) du$$

$$= \left[3u - \frac{2u^2}{2} \right]_0^3$$

$$\begin{aligned}
 &= [3u - u^2]^3 \\
 &= [3(3) - (3)^2] - [3(0) - (0)^2] \\
 &= (9 - 9) - 0 \\
 &= 0 \text{ Ans.}
 \end{aligned}$$

Question: NO: 21

$$\int_0^1 \int_2^{4-2u} dy \, du$$

Solution:

$$\begin{aligned}
 &= \int_0^1 [y]_2^{4-2u} du \\
 &= \int_0^1 (4-2u) - 2 \, du \\
 &= \int_0^1 4-2u-2 \, du \\
 &= \int_0^1 2-2u \, du \\
 &= [2u]_0^1 - \left[\frac{2u^2}{2} \right]_0^1 \\
 &= [2(1) - 2(0)] - [(1)^2 - (0)^2] \\
 &= 2 - 1 \Rightarrow = 1 \text{ Ans.}
 \end{aligned}$$

Question: No: 22

$$\int_0^2 \int_{y-2}^0 du \, dy$$

Solution:

$$\begin{aligned}
 &= \int_0^2 [u]_{y-2}^0 dy \\
 &= \int_0^2 [(0) - (y-2)] dy
 \end{aligned}$$

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$$= \int_0^2 (-y + 2) dy$$

$$= \int_0^2 (2 - y) dy$$

$$= [2y]_0^2 - \left[\frac{y^2}{2} \right]_0^2$$

$$= [2(2) - 2(0)] - \left[\frac{(2)^2}{2} - \frac{(0)^2}{2} \right]$$

$$= (4 - 0) - \left(\frac{4}{2} \right)$$

$$= 4 - \frac{4}{2} \Rightarrow \frac{8 - 4}{2}$$

$$= \frac{4}{2} = 2 \text{ Ans.}$$

Question: NO: 23

$$\int_0^1 \int_y^{\sqrt{y}} dx dy$$

Solution:

$$= \int_0^1 [x]_y^{\sqrt{y}} dy$$

$$= \int_0^1 (\sqrt{y} - y) dy$$

$$= \int_0^1 y^{1/2} dy - \int_0^1 y dy$$

$$= \left[\frac{y^{3/2}}{3/2} \right]_0^1 - \left[\frac{y^2}{2} \right]_0^1$$

$$= \left[\frac{2 y^{3/2}}{3} \right]_0^1 - \left[\frac{(1)^2}{2} - \frac{(0)^2}{2} \right]$$

$$= \left[\frac{2(1)^{3/2}}{3} - \frac{2(0)^{3/2}}{3} \right] - \left[\frac{1}{2} \right]$$

$$= \frac{2}{3} - \frac{1}{2} \Rightarrow \frac{4 - 3}{6} = \frac{1}{6} \text{ Ans.}$$

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Question: No: 24

$$\int_0^1 \int_{1-x}^{1-x^2} dy dx$$

Solution:

$$= \int_0^1 \int_{1-u}^{1-u^2} dy du$$

$$= \int_0^1 [y]_{1-u}^{1-u^2} du$$

$$= \int_0^1 (1-u^2 - (1-u)) du$$

$$= \int_0^1 (1-u^2 - 1 + u) du$$

$$= \int_0^1 (-u^2 + u) du$$

$$= \int_0^1 -u^2 du + \int_0^1 u du$$

$$= -\left[\frac{u^3}{3}\right]_0^1 + \left[\frac{u^2}{2}\right]_0^1$$

$$= -\left[\frac{1^3}{3} - \frac{(0)^3}{3}\right] + \left[\frac{1^2}{2} - \frac{(0)^2}{2}\right]$$

$$= -\frac{1}{3} + \frac{1}{2} \Rightarrow = \frac{-2+3}{6} = \frac{1}{6}$$

Question: No: 25

$$\int_0^1 \int_1^{e^x} dy dx$$

Solution:

$$= \int_0^1 \int_1^{e^x} dy dx$$

$$= \int_0^1 [y]_1^{e^x} dx$$

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$$\int_0^1 [e^u - 1] du$$

$$= \int_0^1 e^u du - \int_0^1 1 du$$
$$= [e^u]_0^1 - [u]_0^1$$

$$= [e^{(1)} - e^{(0)}] - [1 - 0]$$

$$= e - 1 - 1$$

$$= e - 2 \text{ Ans.}$$

Question: NO: 26

$$\int_0^{\ln 2} \int_0^2 e^x dx dy$$

Solution:

$$= \int_0^{\ln 2} [x]_0^2 e^x dy$$

$$= \int_0^{\ln 2} (2 - e^x) dy$$

$$= \int_0^{\ln 2} 2 dy - \int_0^{\ln 2} e^x dy$$

$$= 2[y]_0^{\ln 2} - [e^x]_0^{\ln 2}$$

$$= [2 \ln 2 - 2(0)] - (e^{\ln 2} - e^0)$$

$$= 2 \ln 2 - 2 - 1$$

$$= 2 \ln 2 - 3 \text{ Ans.}$$

Question: No: 27

$$\int_0^{3/2} \int_0^{9-4u^2} 16u \, dy \, du$$

Solution:

$$= \int_0^{3/2} \left[\frac{16u y^2}{2} \right]_0^{9-4u^2} du$$

$$= \int_0^{3/2} [16u y]_0^{9-4u^2} du$$

$$= \int_0^{3/2} [16u(9-4u^2)] - [16u(0)] du$$

$$= \int_0^{3/2} (144u - 64u^3) du$$

$$= \int_0^{3/2} 144u \, du - \int_0^{3/2} 64u^3 \, du$$

$$= 144 \int_0^{3/2} u \, du - 64 \int_0^{3/2} u^3 \, du$$

$$= \left[\frac{144u^2}{2} \right]_0^{3/2} - \left[\frac{64u^4}{4} \right]_0^{3/2}$$

$$= \left[\frac{144 \left(\frac{3}{2} \right)^2}{2} - \frac{144(0)^2}{2} \right] - \left[\frac{16u^4}{1} \right]_0^{3/2}$$

$$= \left[\frac{18}{1} \left(\frac{9}{4} \right) \right] - \left[16 \left(\frac{3}{2} \right)^4 - 16(0)^4 \right]$$

$$= (162) - 16 \left(\frac{81}{16} \right) - 0$$

$$= 162 - 81$$

$$= 81 \text{ Ans.}$$

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Questions NO: 28

$$\int_0^2 \int_0^{4-y^2} y \, du \, dy$$

Solution:

$$= \int_0^2 \int_0^{4-y^2} y \, du \, dy$$

$$= \int_0^2 [xy]_0^{4-y^2} dy$$

$$= \int_0^2 [(4-y^2)y] - [(0)y] dy$$

$$= \int_0^2 (4y - y^3) dy$$

$$= 4 \int_0^2 y \, dy - \int_0^2 y^3 \, dy$$

$$= \left[\frac{4y^2}{2} \right]_0^2 - \left[\frac{y^4}{4} \right]_0^2$$

$$= \left[2y^2 \right]_0^2 - \left[\frac{(2)^4}{4} - \frac{(0)^4}{4} \right]$$

$$= [2(2)^2 - 2(0)^2] - \left(\frac{16}{4} \right)$$

$$= 2(4) - 4$$

$$= 8 - 4 \Rightarrow 4 \text{ Ans.}$$

Question: No: 29

$$\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 3y \, du \, dy$$

Solution:

$$= \int_0^1 \frac{[3xy]^{\sqrt{1-y^2}}}{\sqrt{1-y^2}} dy$$

$$= \int_0^1 [3(\sqrt{1-y^2})y] - [3(-\sqrt{1-y^2})y] dy$$

$$= \int_0^1 3y\sqrt{1-y^2} dy - \int_0^1 -3y\sqrt{1-y^2} dy$$

$$= \int_0^1 3y\sqrt{1-y^2} dy + \int_0^1 3y\sqrt{1-y^2} dy$$

Now by using substitution method

$$\text{Let } u = 1 - y^2$$

$$\frac{du}{dy} = -2y$$

$$du = -2y dy$$

$$y dy = -\frac{du}{2}$$

Now

$$= 3 \int_0^1 y\sqrt{1-y^2} dy + 3 \int_0^1 y\sqrt{1-y^2} dy$$

$$= 3 \int_0^1 (\sqrt{1-y^2}) y dy + 3 \int_0^1 (\sqrt{1-y^2}) y dy$$

$$= 3 \int_0^1 -\sqrt{u} \frac{du}{2} + 3 \int_0^1 -\sqrt{u} \frac{du}{2}$$

$$= -\frac{3}{2} \int_0^1 \sqrt{u} du - \frac{3}{2} \int_0^1 \sqrt{u} du$$

$$= -\frac{3}{2} \int_0^1 (u)^{1/2} du - \frac{3}{2} \int_0^1 (u)^{1/2} du$$

$$= \left[-\frac{3}{2} \cdot \frac{u^{3/2}}{3/2} \right]_0^1 - \left[\frac{3}{2} \cdot \frac{u^{3/2}}{3/2} \right]_0^1$$

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$$= \left[\frac{2}{2} \cdot \frac{2}{3} u^{3/2} \right]_0^1 - \left[\frac{2}{2} \cdot \frac{2}{3} u^{3/2} \right]_0^1$$

$$= \left[-u^{3/2} \right]_0^1 - \left[u^{3/2} \right]_0^1$$

$$= \left[-\frac{2}{3} u^{3/2} \right]_0^1$$
 Now, put the value of u , so

$$= \left[-(\sqrt{1-y^2})^{3/2} \right]_0^1 - \left[(\sqrt{1-y^2})^{3/2} \right]_0^1$$

$$= \left[-(\sqrt{1-(1)^2})^{3/2} - (\sqrt{1-(0)^2})^{3/2} \right] - \left[(\sqrt{1-(1)^2})^{3/2} - ((\sqrt{1-(0)^2})^{3/2}) \right]$$

$$= -[(\sqrt{0})^{3/2} - (\sqrt{1})^{3/2}] - [(\sqrt{0})^{3/2} - (\sqrt{1})^{3/2}]$$

$$= -(-1) - (-1)$$

$$= 1 + 1 = 2 \text{ Ans.}$$

Question: NO: 30

$$\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6x \, dy \, dx$$

Solution:

$$= \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6x \, dy \, dx$$

$$= \int_0^2 6x \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dy \, dx$$

$$= \int_0^2 [6xy]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

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$$= \int_0^2 [6u(\sqrt{4-u^2}) - 6u(-\sqrt{4-u^2})] du$$

$$= \int_0^2 (\sqrt{4-u^2}) u du + 6 \int_0^2 (\sqrt{4-u^2}) u du$$

Now, by using substitution method

Let

$$u = 4 - u^2$$

$$\frac{du}{du} = -2u$$

$$du = -2u du$$

$$u du = -\frac{du}{2}$$

Now,

$$= 6 \int_0^2 \sqrt{4-u^2} \cdot \frac{du}{2} + 6 \int_0^2 \sqrt{4-u^2} \cdot \frac{du}{2}$$

$$= \frac{6}{2} \int_0^2 \sqrt{4-u^2} du + \frac{6}{2} \int_0^2 \sqrt{4-u^2} du$$

$$= 3 \int_0^2 (u)^{1/2} du + 3 \int_0^2 (u)^{1/2} du$$

$$= \left[3 \frac{u^{3/2}}{3/2} \right]_0^2 + \left[3 \frac{u^{3/2}}{3/2} \right]_0^2$$

$$= \left[\frac{3 \cdot 2 u^{3/2}}{3} \right]_0^2 + \left[\frac{3 \cdot 2 u^{3/2}}{3} \right]_0^2$$

$$= \left[2 u^{3/2} \right]_0^2 + \left[2 u^{3/2} \right]_0^2$$

Put the value of u , so

$$= \left[2 (\sqrt{4-u^2})^{3/2} \right]_0^2 + \left[2 (\sqrt{4-u^2})^{3/2} \right]_0^2$$

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$$= \left[2(\sqrt{4 - (2)^2})^{3/2} - 2(\sqrt{4 - (0)^2})^{3/2} \right] - \left[2(\sqrt{4 - (2)^2})^{3/2} - 2(\sqrt{4 - (0)^2})^{3/2} \right]$$

$$= \left[2(\sqrt{4-4})^{3/2} - 2(\sqrt{4})^{3/2} \right] - \left[2(\sqrt{4-4})^{3/2} - 2(\sqrt{4})^{3/2} \right]$$

$$= \left[2(\sqrt{0})^{3/2} - 2(2)^{3/2} \right] - \left[2(\sqrt{0})^{3/2} - 2(2)^{3/2} \right]$$

$$= \left[0 - (4)^{3/2} \right] - \left[0 - (4)^{3/2} \right]$$

$$= -(-4^{3/2}) - (-4^{3/2})$$

$$= 4^{3/2} + 4^{3/2}$$

$$= (2^2)^{3/2} + (2^2)^{3/2}$$

$$= 2^{2 \times \frac{3}{2}} + 2^{2 \times \frac{3}{2}}$$

$$= 2^3 + 2^3$$

$$= 8 + 8 \Rightarrow = 16 \text{ Ans.}$$

Topic:-

Normal vector to the
Surface

$$n = \frac{\partial r}{\partial u} + \frac{\partial r}{\partial v} \quad \text{OR} \quad n = \frac{\partial r}{\partial \theta} + \frac{\partial r}{\partial \phi}$$

Here $r = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

Question: NO: 01 $\Rightarrow$ Find the surface

$$x = 2 \cosh u \cos v \quad ; \quad y = 3 \cosh u \sin v$$

$z = 6 \sinh u$. Find a normal
vector to the surface for which

$$v = \frac{\pi}{3}.$$

Solution:

$$r = (2 \cosh u \cos v)\mathbf{i} + (3 \cosh u \sin v)\mathbf{j} + (6 \sinh u)\mathbf{k}$$

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$$\frac{dr}{du} = (2 \sinh u \cos v) i + (3 \sinh u \sin v) j + (6 \cosh u) k$$

$$\left. \frac{dr}{du} \right|_{v=\pi/3} = [2 \sinh u \cos(\frac{\pi}{3})] i + [3 \sinh u \sin(\frac{\pi}{3})] j + (6 \cosh u) k$$

$$\left. \frac{dr}{du} \right|_{v=\pi/3} = [2 \sinh u \cdot \frac{1}{2}] i + [3 \sinh u (\frac{\sqrt{3}}{2})] j + (6 \cosh u) k$$

$$\frac{dr}{du} = \sinh u i + \frac{3\sqrt{3}}{2} \sinh u j + 6 \cosh u k \rightarrow (1)$$

Now,

$$\frac{dr}{dv} = [2 \cosh u (-\sin v)] i + [3 \cosh u \cos v] j + 0 k$$

$$\frac{dr}{dv} = (-2 \cosh u \sin v) i + (3 \cosh u \cos v) j + 0 k$$

$$\left. \frac{dr}{dv} \right|_{v=\pi/3} = [-2 \cosh u \sin(\frac{\pi}{3})] i + [3 \cosh u \cos(\frac{\pi}{3})] j + 0 k$$

$$\left. \frac{dr}{dv} \right|_{v=\pi/3} = [-2 \cosh u (\frac{\sqrt{3}}{2})] i + [3 \cosh u \cdot \frac{1}{2}] j + 0 k$$

$$\frac{dr}{dv} = -\cosh u \sqrt{3} i + \frac{3}{2} \cosh u j + 0 k \rightarrow (2)$$

Now,

	i	j	k
$n = \frac{dr}{du} \times \frac{dr}{dv}$	$\sinh u$	$\frac{3\sqrt{3}}{2} \sinh u$	$6 \cosh u$
	$-\sqrt{3} \cosh u$	$\frac{3}{2} \cosh u$	0

$$= i(0 - \frac{3}{2} \cdot 3 \cosh^2 u) - j(0 + 6\sqrt{3} \cosh^2 u) + k[(\sinh u \cdot \frac{3}{2} \cosh u) - (-\sqrt{3} \cosh u \cdot \frac{3\sqrt{3}}{2} \sinh u)]$$

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$$\begin{aligned}
 &= 9 \cosh^2 u i - 6\sqrt{3} \cosh^2 u j + \\
 &k \left[\frac{3}{2} \sinh u \cosh u + 3(\sqrt{3})^2 \sinh u \cosh u \right] \\
 &= 9 \cosh^2 u i - 6\sqrt{3} \cosh^2 u j + \left(\frac{3}{2} + 9 \right) \sinh u \cosh u k \\
 &= 9 \cosh^2 u i - 6\sqrt{3} \cosh^2 u j + \left(\frac{3+9}{2} \right) \sinh u \cosh u k \\
 &= 9 \cosh^2 u i - 6\sqrt{3} \cosh^2 u j + \frac{12}{2} \sinh u \cosh u k \\
 &= 9 \cosh^2 u i - 6\sqrt{3} \cosh^2 u j + 6 \sinh u \cosh u k \rightarrow \text{Ans.}
 \end{aligned}$$

Question: NO: 02 $\Rightarrow$ Find the surface

$$x = (3 + \cos \phi) \cos \theta; \quad y = (3 + \cos \phi) \sin \theta;$$

$z = \sin \phi$. Find a normal vector to

surface for which $\theta = \pi/4; \quad \phi = \frac{2\pi}{3}$.

Solution:

$$\begin{aligned}
 \mathbf{r} &= \frac{\partial \mathbf{r}}{\partial \theta} \times \frac{\partial \mathbf{r}}{\partial \phi} \\
 \mathbf{r} &= [(3 + \cos \phi) \cos \theta] i + [(3 + \cos \phi) \sin \theta] j \\
 &\quad + (\sin \phi) k
 \end{aligned}$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = [(3 + \cos \phi) (-\sin \theta)] i + [(3 + \cos \phi) \cos \theta] j + 0k$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = [(3 + \cos \phi) \sin \frac{\pi}{4}] i + [(3 + \cos \phi) \cos \frac{\pi}{4}] j + 0k$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = -[(3 + \cos \frac{2\pi}{3})] \left(\frac{1}{\sqrt{2}} \right) i + [(3 + \cos \frac{2\pi}{3})] \left(\frac{1}{\sqrt{2}} \right) j + 0k$$

$$\frac{\partial \mathbf{r}}{\partial \theta} = -\left[\left(3 + \frac{1}{2} \right) \left(\frac{1}{\sqrt{2}} \right) \right] i + \left[\left(3 - \frac{1}{2} \right) \left(\frac{1}{\sqrt{2}} \right) \right] j + 0k$$

$$\frac{\partial \mathbf{r}}{\partial \theta}$$

$$\frac{\partial \mathbf{r}}{\partial \theta}$$

$$\frac{\partial \mathbf{r}}{\partial \theta}$$

Now,

$$\frac{\partial \mathbf{r}}{\partial \phi}$$

$$\frac{\partial \mathbf{r}}{\partial \phi} = (-$$

$$\frac{\partial \mathbf{r}}{\partial \phi} = -$$

$$\frac{\partial \mathbf{r}}{\partial \phi} = -$$

$$\frac{\partial \mathbf{r}}{\partial \phi}$$

Now,

$$\begin{aligned}
 &= \frac{5\sqrt{2}}{4} \\
 &= \frac{\sqrt{3}}{2\sqrt{2}}
 \end{aligned}$$

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$$\frac{\partial r}{\partial \theta} = - \left[\left(\frac{6-1}{2} \right) \left(\frac{1}{\sqrt{2}} \right) \right] \underline{i} + \left[\left(\frac{6-1}{2} \right) \left(\frac{1}{\sqrt{2}} \right) \right] \underline{j} + 0 \underline{k}$$

$$\frac{\partial r}{\partial \theta} = - \left(\frac{5}{2\sqrt{2}} \right) \underline{i} + \left(\frac{5}{2\sqrt{2}} \right) \underline{j} + 0 \underline{k} \rightarrow (1)$$

$$\frac{\partial r}{\partial \theta} = - \frac{5\sqrt{2}}{4} \underline{i} + \frac{5\sqrt{2}}{4} \underline{j} + 0 \underline{k} \rightarrow (1)$$

Now,

$$\frac{\partial r}{\partial \phi} = [\{0 + (-\sin \phi)\} \cos \theta] \underline{i} + [\{0 + (-\sin \phi)\} \sin \theta] \underline{j} + \cos \phi \underline{k}$$

Putting the values, so get

$$\frac{\partial r}{\partial \phi} = (-\sin \phi \cos \theta) \underline{i} + (-\sin \phi \sin \theta) \underline{j} + \cos \phi \underline{k}$$

$$\frac{\partial r}{\partial \phi} = - \left(\sin \frac{2\pi}{3} \cdot \cos \frac{\pi}{4} \right) \underline{i} - \left(\sin \frac{2\pi}{3} \cdot \sin \frac{\pi}{4} \right) \underline{j} + \cos \frac{2\pi}{3} \underline{k}$$

$$\frac{\partial r}{\partial \phi} = - \left(\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \right) \underline{i} - \left(\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} \right) \underline{j} + \left(-\frac{1}{2} \right) \underline{k}$$

$$\frac{\partial r}{\partial \phi} = - \frac{\sqrt{3}}{2\sqrt{2}} \underline{i} - \frac{\sqrt{3}}{2\sqrt{2}} \underline{j} - \frac{1}{2} \underline{k} \rightarrow (2)$$

Now,

$$n = \frac{\partial r}{\partial \theta} \times \frac{\partial r}{\partial \phi}$$

$$n = \begin{vmatrix} i & j & k \\ -\frac{5}{2\sqrt{2}} & \frac{5}{2\sqrt{2}} & 0 \\ -\frac{\sqrt{3}}{2\sqrt{2}} & -\frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{2} \end{vmatrix}$$

$$n = i \left[\left(\frac{5}{2\sqrt{2}} \right) \left(-\frac{1}{2} \right) - \left(-\frac{\sqrt{3}}{2\sqrt{2}} \right) (0) \right] - j \left[\left(-\frac{5}{2\sqrt{2}} \right) \left(-\frac{1}{2} \right) - \left(-\frac{\sqrt{3}}{2\sqrt{2}} \right) (0) \right] + k \left[\left(-\frac{5}{2\sqrt{2}} \right) \left(-\frac{\sqrt{3}}{2\sqrt{2}} \right) - \left(\frac{5}{2\sqrt{2}} \right) \left(-\frac{\sqrt{3}}{2\sqrt{2}} \right) \right]$$

$$n = \left(\frac{5}{4\sqrt{2}} - 0 \right) i - j \left(\frac{5}{4\sqrt{2}} - 0 \right) + k \left[\frac{5\sqrt{3}}{2(\sqrt{2})^2} + \frac{5\sqrt{3}}{4(\sqrt{2})^2} \right]$$

$$n = -\frac{5}{4\sqrt{2}} i - \frac{5}{4\sqrt{2}} j + \left(\frac{5\sqrt{3}}{8} + \frac{5\sqrt{3}}{8} \right) k$$

$$n = -\frac{5}{4\sqrt{2}} i - \frac{5}{4\sqrt{2}} j + \left(\frac{2 \cdot 5\sqrt{3}}{8} \right) k$$

$$n = -\frac{5}{4\sqrt{2}} i - \frac{5}{4\sqrt{2}} j + \frac{5\sqrt{3}}{4} k \rightarrow \text{Ans}$$